



**International
Standard**

**ISO/IEC/IEEE
15026-1**

**Systems and software
engineering — Systems and
software assurance —**

**Part 1:
Vocabulary and concepts**

*Ingénierie des systèmes et du logiciel — Assurance des systèmes
et du logiciel —*

Partie 1: Vocabulaire et concepts

**Second edition
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Foreword

ISO (the International Organization for Standardization) and IEC (the International Electrotechnical Commission) form the specialized system for worldwide standardization. National bodies that are members of ISO or IEC participate in the development of International Standards through technical committees established by the respective organization to deal with particular fields of technical activity. ISO and IEC technical committees collaborate in fields of mutual interest. Other international organizations, governmental and non-governmental, in liaison with ISO and IEC, also take part in the work.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives or www.iec.ch/members_experts/refdocs).

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For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see www.iso.org/iso/foreword.html. In the IEC, see www.iec.ch/understanding-standards.

This document was prepared by Joint Technical Committee ISO/IEC JTC 1, *Information technology*, Subcommittee SC 7, *Software and systems engineering*, in cooperation with the Systems and Software Engineering Standards Committee of the IEEE Computer Society, under the Partner Standards Development Organization cooperation agreement between ISO and IEEE.

This second edition cancels and replaces the first edition (ISO/IEC/IEEE 15026-1:2019), which has been technically revised.

The main changes are as follows:

- definitions of terms introduced in other parts of the ISO/IEC/IEEE 15026 series have been added or modified;
- definitions of terms whose definitions were sourced from ISO/IEC 15288 and ISO/IEC/IEEE 24774 have been updated.

A list of all parts in the ISO/IEC/IEEE 15026 series can be found on the ISO and IEC websites.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html and www.iec.ch/national-committees.

Introduction

Software and systems assurance and closely related fields share concepts but have different vocabularies and perspectives. This document provides an unambiguous use of vocabulary for systems and software assurance and a unifying set of underlying concepts across these various fields. It provides a basis for elaboration, discussion and recording agreement and rationale regarding concepts and the vocabulary used uniformly across the ISO/IEC/IEEE 15026 series.

[Clause 4](#) covers basic concepts such as assurance, stakeholders, systems and products, property, uncertainty and confidence, conditions and initial events, and consequence. [Clause 5](#) covers some issues of which users of ISO/IEC/IEEE 15026-2, ISO/IEC/IEEE 15026-3 and ISO/IEC/IEEE 15026-4 should be initially aware ([5.2](#)). [Clause 6](#), [Clause 7](#) and [Clause 8](#) cover concepts relevant to users of ISO/IEC/IEEE 15026-2, ISO/IEC/IEEE 15026-3 and ISO/IEC/IEEE 15026-4, respectively; also, users of one of these parts can benefit from the clauses for other parts.

The essential concepts introduced by the ISO/IEC/IEEE 15026 series are the claims in an assurance case and the support of claims in terms of argument and evidence. These claims are in the context of assurance for properties of systems and software within life cycle processes for the system or software product.

Potential users of the ISO/IEC/IEEE 15026 series are developers and maintainers of assurance cases and those who wish to develop, sustain, evaluate or acquire a system that possesses requirements for specific properties in such a way as to be more certain of those properties and their requirements. The ISO/IEC/IEEE 15026 series uses concepts and vocabulary consistent with ISO/IEC/IEEE 12207 and ISO/IEC/IEEE 15288 and generally consistent with the standards on Systems and software Quality Requirements and Evaluation (SQuaRE) developed by JTC 1/SC 7, but the concepts and vocabulary provided by the ISO/IEC/IEEE 15026 series can differ from those to which the potential user is accustomed. This document attempts to clarify these differences.

The ISO/IEC/IEEE 15026 series is made up of the following parts.

- ISO/IEC/IEEE 15026-1 explains concepts and terms as a basis for all parts of the ISO/IEC/IEEE 15026 series.
- ISO/IEC/IEEE 15026-2 includes requirements on the structure of the assurance case.
- ISO/IEC/IEEE 15026-3 relates integrity levels to the assurance case and includes requirements for their use with and without an assurance case.
- ISO/IEC/IEEE 15026-4 provides guidance and recommendations for assurance of a selected claim about the system-of-interest by achieving the claim and showing the achievement. The guidance and recommendations are given in a system assurance process view on top of ISO/IEC/IEEE 15288 and a software assurance process view on top of ISO/IEC/IEEE 12207.

The assurance case is relevant to a greater or lesser extent in all parts of the ISO/IEC/IEEE 15026 series, although ISO/IEC/IEEE 15026-4 discusses achieving the claim and showing the achievement of the claim whether or not such “showing” is contained in an artefact specifically called an “assurance case”.

ISO/IEC/IEEE 15026-2 concentrates on the contents and structure of the assurance case. ISO/IEC/IEEE 15026-3 relates integrity levels and assurance cases by describing how integrity levels and assurance cases can work together, especially in the definition of specifications for integrity levels or by using integrity levels within a portion of an assurance case. This relationship is governed by the degree of risk and dependencies in the system.

ISO/IEC/IEEE 15026-4 includes assurance-related guidance and recommendations for activities across the life cycle processes including activities that extend beyond those directly related to an assurance case, e.g. project planning for assurance-related considerations.

Systems and software engineering — Systems and software assurance —

Part 1: Vocabulary and concepts

1 Scope

This document defines assurance-related terms and establishes an organized set of concepts to form a basis for shared understanding in the field of assurance. It benefits users of ISO/IEC/IEEE 15026-2, ISO/IEC/IEEE 15026-3 and ISO/IEC/IEEE 15026-4.

Vocabulary and concepts for assurance of a service being operated and managed on an ongoing basis is not covered in this document.

While essential to assurance practice, details regarding exactly how to measure, demonstrate or analyse particular properties are not covered.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

ISO, IEC and IEEE maintain terminology databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <https://www.electropedia.org/>
- IEEE Standards Dictionary Online: available at <https://ieeexplore.ieee.org/xpls/dictionary.jsp>

NOTE For additional terms and definitions in the field of systems and software engineering, see ISO/IEC/IEEE 24765, which is published periodically as a “snapshot” of the SEVOCAB (Systems and software engineering vocabulary) database and is publicly accessible at <https://www.computer.org/sevocab>.

3.1 Terms related to assurance

3.1.1

assurance

grounds for justified confidence that a *claim* ([3.1.10](#)) has been or will be achieved

Note 1 to entry: By definition, assurance is about a claim.

Note 2 to entry: The claim can be a conjunction of more than one claim.

3.1.2

assurance argument

artefact that links tangible *evidence* (3.1.15) and assumptions to provide a convincing and valid argument of a *claim* (3.1.10) under a given context

Note 1 to entry: An assurance argument provides the reasoning that connects the evidence (and assumptions) to the claim. It explains 'why' the evidence supports the claim and how the evidence is relevant.

Note 2 to entry: An argument is valid only if all of its premises being true implies that its conclusion is true.

3.1.3

assurance case

auditable artefact that provides a convincing and sound argument for a *claim* (3.1.10) on the basis of tangible *evidence* (3.1.15) under a given context

Note 1 to entry: Premises of an argument are propositions, and all propositions are either true or false.

Note 2 to entry: An argument is sound only if it is valid and contains only true premises.

Note 3 to entry: An argument is not sound if it is valid but contains some false premises.

3.1.4

assurance case report

auditable artefact that provides the *claim* (3.1.10) of an *assurance case* (3.1.3) and a complete index of the argument and *evidence* (3.1.15) of the assurance case such that the assurance case of interest can be assembled from the index

Note 1 to entry: An assurance case report is mapped to the report field. An assurance case report is an introduction to the assurance case. Its content varies from one case to another. For simple assurance cases, a report can be superfluous. For larger or more complex assurance cases the report can include a *system* (3.2.8) description, some of the argument and its supporting evidence e.g. the higher-level arguments and their supporting evidence.

Note 2 to entry: Assurance case reports are often issued as periodic updates to the assurance case at pre-agreed points such as at the end of a life cycle stage. They report on results of assurance activities done so far, an assessment of overall progress and a review of assurance activities' performance. Assurance case reports can be used at decision gates for stage exit/entry by suppliers and for project monitoring by acquirers.

3.1.5

assurance claim

claim (3.1.10) for which *assurance* (3.1.1) is considered

3.1.6

assurance information

information including a *claim* (3.1.10) about a *system* (3.2.8), *evidence* (3.1.15) supporting the claim, an argument showing how the evidence supports the achievement of the claim, and the context for these items

Note 1 to entry: The sub-claims included in the argument of assurance information can be about the life cycle of the *system-of-interest* (3.2.10) when, for example, the top-level claim implies continuous achievement of some *property* (3.1.17).

Note 2 to entry: ISO/IEC/IEEE 15026-2 specifies *assurance cases* (3.1.3) that documents assurance information.

3.1.7

assurance objective

purpose of achievement of the *assurance claim* (3.1.5)

Note 1 to entry: Assurance objectives determine the required degree of *integrity level* (3.3.1) and permissible uncertainty in the *assurance information* (3.1.6).

3.1.8

attribute

inherent *property* ([3.1.17](#)) or characteristic of an entity that can be distinguished quantitatively or qualitatively by human or automated means

Note 1 to entry: ISO 9000 distinguishes two types of attributes: a permanent characteristic existing inherently in something; and an assigned characteristic of a *product* ([3.2.6](#)), *process* ([3.2.4](#)), or *system* ([3.2.8](#)) (e.g. the price of a product, the owner of a product).

[SOURCE: ISO/IEC/IEEE 29148:2018, 3.1.2]

3.1.9

basic assumption

assumption in a collection of *claims* ([3.1.10](#)) shared by *assurance cases* ([3.1.3](#)) in a specific field

Note 1 to entry: Such a collection corresponds to a set of axioms in formal system in logic.

3.1.10

claim

true-false statement about the limitations on the values of an unambiguously defined *property* ([3.1.17](#)) and limitations on the uncertainty of the property's values falling within these limitations during the claim's duration of applicability under stated *conditions* ([3.1.11](#)).

Note 1 to entry: The unambiguously defined property is called the claim's property.

Note 2 to entry: Uncertainties may also be associated with the duration of applicability and the stated conditions.

Note 3 to entry: A claim potentially contains the following:

- property of the *system-of-interest* ([3.2.10](#));
- limitations on the value of the property associated with the claim (e.g. on its range);
- limitations on the uncertainty of the property value meeting its limitations;
- limitations on duration of claim's applicability;
- duration-related uncertainty;
- limitations on conditions associated with the claim;
- condition-related uncertainty.

Note 4 to entry: The term “limitations” is used to fit the many situations that can exist. Values can be a single value or multiple single values, a range of values or multiple ranges of values, and can be multi-dimensional. The boundaries of these limitations are sometimes not sharp, e.g., they can involve probability distributions and can be incremental.

3.1.11

condition

measurable qualitative or quantitative *attribute* ([3.1.8](#)) that is stipulated for a *requirement* ([3.2.7](#)) and that indicates a circumstance or event under which a requirement applies

[SOURCE: ISO/IEC/IEEE 29148:2018, 3.1.6]

3.1.12

constraint

externally imposed limitation on the *system* ([3.2.8](#)), its design, or implementation or on the *process* ([3.2.4](#)) used to develop or modify a system

Note 1 to entry: A constraint is a factor that is imposed on the solution by force or compulsion and may limit or modify the design.

[SOURCE: ISO/IEC/IEEE 29148:2018, 3.1.7]

3.1.13

critical property

property ([3.1.17](#)) that is agreed by primary stakeholders as having serious *consequence* ([3.4.3](#))

3.1.14

dependability

<of an item> ability to perform as and when required

Note 1 to entry: Dependability includes availability, reliability, recoverability, maintainability, and maintenance support performance, and, in some cases, other characteristics such as durability, safety and security.

Note 2 to entry: Dependability is used as a collective term for the time-related quality characteristics of an item.

[SOURCE: IEC 60050-192:2015, 192-01-22]

3.1.15

evidence

data referred to by an argument that validates the given claim

Note 1 to entry: Evidence can include result of tests, inspections, simulations, audits.

Note 2 to entry: The role of the evidence is to support a demonstrable argument that the claim is valid.

3.1.16

inference

reasoning step that derives a *claim* ([3.1.10](#)) from a list of claims (premises) under a specified context

Note 1 to entry: The claim derived by inference is the conclusion, and the supported claims from which the inference derives the conclusion are the premises.

Note 2 to entry: There are special inferences called obvious inferences. See 5.3.4 EXAMPLE 2.

3.1.17

property

attribute ([3.1.8](#)) of things

3.1.18

type

set of values, elements or items

3.1.19

undeveloped argument

placeholder argument to be replaced by a concrete argument

3.2 Terms related to system life cycles

3.2.1

approval authority

person (or persons) or *organization* ([3.2.3](#)) (or organizations), or both, responsible for approving activities, artefacts and other aspects of the *system* ([3.2.8](#)) during its life cycle

Note 1 to entry: The approval authority may include multiple entities, e.g. individuals or organizations. These can include different entities with either different levels of approval or different areas of interest, or both.

Note 2 to entry: In two-party situations, the approval authority often rests with the acquirer. In regulatory situations, the approval authority may be a third party such as a governmental organization or its agent. In other situations, for example, the purchase of off-the-shelf *products* ([3.2.6](#)) developed by a single-party, the independence of the approval authority can be a relevant issue to the acquirer.

3.2.2

design authority

person or *organization* ([3.2.3](#)) that is responsible for the design of the *product* ([3.2.6](#))

3.2.3

organization

person or group of people that has its own functions with responsibilities, authorities, and relationships to achieve its objectives

EXAMPLE Company, corporation, firm, enterprise, manufacturer, institution, charity, sole trader, association, or parts or combination thereof.

[SOURCE: ISO/IEC/IEEE 15288:2023, 3.25]

3.2.4

process

set of interrelated or interacting activities that transforms inputs into outputs

[SOURCE: ISO/IEC/IEEE 15288:2023, 3.27]

3.2.5

process view

description of how a specified purpose and set of outcomes can be achieved by employing the activities and tasks of existing *processes* ([3.2.4](#))

[SOURCE: ISO/IEC/IEEE 24774:2021, 3.15]

3.2.6

product

output of an *organization* ([3.2.3](#)) that can be produced without any transaction taking place between the organization and the customer

Note 1 to entry: The dominant element of a product is that it is generally tangible.

[SOURCE: ISO 9000:2015, 3.7.6, modified — Notes 1 and 3 to entry have been removed.]

3.2.7

requirement

statement which translates or expresses a need and its associated constraints and *conditions* ([3.1.11](#))

Note 1 to entry: Requirements exist at different levels in the *system* ([3.2.8](#)) structure.

Note 2 to entry: A requirement is an expression of one or more particular needs in a very specific, precise and unambiguous manner.

Note 3 to entry: A requirement always relates to a system, software or service, or other item of interest.

[SOURCE: ISO/IEC/IEEE 29148:2018, 3.1.19]

3.2.8

system

arrangement of parts or elements that together exhibit a stated behaviour or meaning that the individual constituents do not

Note 1 to entry: A system is sometimes considered as a *product* ([3.2.6](#)) or as the services it provides.

Note 2 to entry: In practice, the interpretation of its meaning is frequently clarified by the use of an associative noun, e.g. aircraft system. Alternatively, the word "system" is substituted simply by a context-dependent synonym (e.g. aircraft), though this potentially obscures a system principles perspective.

Note 3 to entry: A complete system includes all of the associated equipment, facilities, material, computer programs, firmware, technical documentation, services and personnel required for operations and support to the degree necessary for self-sufficient use in its intended environment.

[SOURCE: ISO/IEC/IEEE 15288:2023, 3.46]

3.2.9

system element

discrete part of a *system* (3.2.8) that can be implemented to fulfil specified *requirements* (3.2.7)

EXAMPLE Hardware, software, data, humans, *processes* (3.2.4) [e.g. processes for providing service to users], procedures [e.g. operator instructions], facilities, materials, and naturally occurring entities or any combination.

[SOURCE: ISO/IEC/IEEE 15288:2023, 3.47]

3.2.10

system-of-interest

system (3.2.8) whose life cycle is under consideration

[SOURCE: ISO/IEC/IEEE 15288:2023, 3.48, modified — The abbreviated term "SoS" has been removed.]

3.3 Terms related to integrity level

3.3.1

integrity level

degree of confidence that the *system-of-interest* (3.2.10) meets the associated *integrity level claim* (3.3.3)

Note 1 to entry: A definition of "integrity" consistent with its use in "integrity level" has not been agreed in the relevant communities. Hence, no separate definition of "integrity" is included in this document.

Note 2 to entry: An integrity level is different from the *likelihood* (3.4.11) that the integrity level claim is met but they are closely related.

Note 3 to entry: The word "confidence" implies that the definition of integrity levels can be a subjective concept.

Note 4 to entry: In this document, integrity levels are defined in terms of *risk* (3.4.14) and hence cover safety, security, economic and any other dimension of risk that is relevant to the system-of-interest.

3.3.2

integrity level assurance authority

independent person or *organization* (3.2.3) responsible for certifying compliance with the *integrity level requirements* (3.3.5)

3.3.3

integrity level claim

proposition representing a *requirement* (3.2.7) on a *risk reduction measure* (3.4.16) identified in the *risk treatment* (3.4.18) *process* (3.2.4) of the *system-of-interest* (3.2.10)

Note 1 to entry: In general, it is described in terms of requirements to avoid, control or mitigate the *consequences* (3.4.3) of *dangerous conditions* (3.4.4), so as to provide a *tolerable risk* (3.4.21) if it is met.

Note 2 to entry: In the IEC 61508 series, the proposition that an E/E/PE safety-related *system* (3.2.8) satisfactorily performs the specified safety functions under all stated *conditions* (3.1.11) can be regarded as an integrity level claim.

3.3.4

integrity level definition authority

person or *organization* (3.2.3) responsible for defining *integrity levels* (3.3.1) and *integrity level requirements* (3.3.5)

3.3.5

integrity level requirements

set of *requirements* (3.2.7) that, when met, will provide a level of confidence in the associated *integrity level claim* (3.3.3) commensurate with the associated *integrity level* (3.3.1)

3.4 Terms related to risks and dependability

3.4.1

adverse consequence

consequence (3.4.3) that results in a specified level of loss

Note 1 to entry: An adverse consequence results from the *system-of-interest* (3.2.10) being in a *dangerous condition* (3.4.4) combined with the environment of the *system* (3.2.8) being in its worst-case state.

Note 2 to entry: Harm in ISO/IEC Guide 51 is an instance of an adverse consequence. The concept of adverse consequences is introduced in order to cover not only harms in the safety context but also loss of assets in the security context and any other losses.

3.4.2

attack

malicious action or interaction with the *system* (3.2.8) or its environment that has the potential to result in a *fault* (3.4.8) or an *error* (3.4.6), and thereby possibly in a *failure* (3.4.7), or an *adverse consequence* (3.4.1)

3.4.3

consequence

outcome of an event affecting objectives

[SOURCE: ISO 31073:2022, 3.3.18, modified — Notes 1, 2 and 3 to entry have been removed.]

3.4.4

dangerous condition

state of a *system* (3.2.8) that, in combination with some states of the environment, will result in an *adverse consequence* (3.4.1)

Note 1 to entry: A hazardous situation in ISO/IEC Guide 51 and IEC 61508-4 can be a dangerous condition. A threat in the ISO/IEC/IEEE 15026 series is also an example of a dangerous condition. A concept of dangerous conditions is introduced in order to cover not only hazardous situations in the safety context but also *errors* (3.4.6) in the reliability, integrity, confidentiality or *dependability* (3.1.14) contexts and other states of a system which can lead to adverse consequences.

Note 2 to entry: Occurrences of *failures* (3.4.7) in the context of reliability or as defined in IEC 61508-4 often but does not always lead to dangerous conditions.

Note 3 to entry: A dangerous condition therefore has at least the following *attributes* (3.1.8):

- a) the associated adverse consequences;
- b) the trigger events that lead to the dangerous condition;
- c) the trigger events that lead to the adverse consequences from the dangerous condition.

3.4.5

desirable consequence

positive consequence

consequence (3.4.3) associated with a benefit or gain or avoiding an *adverse consequence* (3.4.1)

3.4.6

error

discrepancy between a computed, observed or measured value or *condition* (3.1.11), and the true, specified or theoretically correct value or condition

[SOURCE: IEC 60050-192:2015, 192-03-02, modified — Notes 1 and 2 to entry have been removed.]

3.4.7

failure

<of an item> loss of ability to perform as required

Note 1 to entry: A failure of an item is an event that results in a *fault* (3.4.8) of that item.

Note 2 to entry: Qualifiers, such as catastrophic, critical, major, minor, marginal and insignificant, can be used to categorize failures according to the severity of *consequences* (3.4.3), the choice and definitions of severity criteria depending upon the field of application.

Note 3 to entry: Qualifiers, such as misuse, mishandling and weakness, can be used to categorize failures according to the cause of failure.

[SOURCE: IEC 60050-192:2015, 192-03-01]

3.4.8

fault

<of an item> inability to perform as required, due to an internal state

Note 1 to entry: A fault of an item results from a *failure* (3.4.7), either of the item itself, or from a deficiency in an earlier stage of the life cycle, such as specification, design, manufacture or maintenance.

Note 2 to entry: Qualifiers, such as specification, design, manufacture, maintenance or misuse, may be used to indicate the cause of a fault.

Note 3 to entry: The type of fault may be associated with the type of associated failure, e.g. wear-out fault and wear-out failure.

Note 4 to entry: The adjective “faulty” designates an item having one or more faults.

[SOURCE: IEC 60050-192:2015, 192-04-01, modified — The cross-reference to “latent fault” in note 1 to entry has been removed.]

3.4.9

initial risk

estimated *risk* (3.4.14) before applying *risk reduction measures* (3.4.16)

3.4.10

level of risk

magnitude of a *risk* (3.4.14) or combination of risks, expressed in terms of the combination of *consequences* (3.4.3) and their *likelihood* (3.4.11)

[SOURCE: ISO 31073:2022, 3.3.22]

3.4.11

likelihood

chance of something happening

[SOURCE: ISO 31073:2022, 3.3.16, modified — Notes 1 and 2 have been removed.]

3.4.12

property-of-interest

object whose loss is considered as a negative effect

Note 1 to entry: The concept of property-of-interest is introduced in order to characterize negative effects of *consequences* (3.4.3).

Note 2 to entry: In the safety context, human lives and health can be property-of-interests.

Note 3 to entry: Assets in the security context, e.g. defined in ISO/IEC 15408-1, can be a property-of-interest.

3.4.13

residual risk

risk (3.4.14) remaining after *risk treatment* (3.4.18)

[SOURCE: ISO 31073:2022, 3.3.38, modified — Notes 1 and 2 have been removed.]

3.4.14**risk**

effect of uncertainty on objectives

Note 1 to entry: An effect is a deviation from the expected. It can be— positive, negative or both. In this document the focus is on negative deviations leading to *adverse consequences* (3.4.1).

Note 2 to entry: Risk is often characterized by reference to potential events and *consequences* (3.4.3), or a combination of these.

Note 3 to entry: Risk is often expressed in terms of a combination of the consequences of an event (including changes in circumstances) and the associated *likelihood* (3.4.11) of occurrence. In this document risk is characterized as the combination of the severity of the adverse consequence and the likelihood of an adverse consequence occurring.

Note 4 to entry: Objectives can have different aspects, such as financial, health and safety, and environmental goals and can apply at different levels, such as strategic, organization-wide, project, *product* (3.2.6) and *process* (3.2.4).

Note 5 to entry: Uncertainty is the state, even partial, of deficiency of information related to, understanding or knowledge of, an event, its consequence, or likelihood.

[SOURCE: ISO 31073:2022, 3.1.1, modified — Notes to entry have been reordered; information specific to this document has been added in Notes 1 and 3 to entry.]

3.4.15**risk criteria**

terms of reference against which the significance of a *risk* (3.4.14) is evaluated

[SOURCE: ISO 31073:2022, 3.3.6, modified — Notes 1 and 2 to entry have been removed.]

3.4.16**risk reduction measure**

measure to reduce or mitigate *risk* (3.4.14)

Note 1 to entry: A typical risk reduction measure is a safety-related *system* (3.2.8) in the IEC 61508 series.

Note 2 to entry: Design, review and test are also risk reduction measures, as are hazards analysis incorporated in the design.

3.4.17**risk source**

element which alone or in combination has the intrinsic potential to give rise to *risk* (3.4.14)

Note 1 to entry: A hazard in ISO 31073 is an instance of a risk source.

Note 2 to entry: A *fault* (3.4.8), an *error* (3.4.6) or a *failure* (3.4.7) in the context of reliability can be a risk source.

Note 3 to entry: A threat in the context of security and a *threat agent* (3.4.20) and an adverse action defined in ISO/IEC 15408-1 can be a risk source.

[SOURCE: ISO 31073:2022, 3.3.10, modified — Notes 1, 2 and 3 to entry have been added.]

3.4.18**risk treatment**

process (3.2.4) to eliminate *risk* (3.4.14) or reduce it to a tolerable level

[SOURCE: ISO 31073:2022, 3.3.32, modified — The words “modify risk” have been replaced with “eliminate risk or reduce it to a tolerable level”; notes 1, 2 and 3 to entry have been removed.]

3.4.19**systematic failure**

failure (3.4.7) that consistently occurs under particular *conditions* (3.1.11) of handling, storage or use

Note 1 to entry: A systematic failure can be reproduced by deliberately applying the same conditions, although not all reproducible failures are systematic.

Note 2 to entry: The cause of a systematic failure originates in the specification, design, manufacture, installation, operation or maintenance of the item.

[SOURCE: IEC 60050-192:2015, 192-03-10]

3.4.20

threat agent

entity that can adversely act on *property-of-interest* ([3.4.12](#))

3.4.21

tolerable risk

level of risk ([3.4.10](#)) which is accepted in a given context based on the current values of society

Note 1 to entry: A tolerable risk is sometimes called an acceptable risk, for example, see ISO/IEC/IEEE 16085 and ISO 14971. The general risk management standards ISO 31073 and ISO 31000 use both phrases without explicit definitions.

[SOURCE: ISO/IEC Guide 51:2014, 3.15, modified — Note 1 to entry has been replaced by a new one.]

3.4.22

violation

behaviour, act or event deviating from a *system's* ([3.2.8](#)) desired property or *claim* ([3.1.10](#)) of interest

Note 1 to entry: In the area of safety, the term “violation” is used to refer to a deliberate human contravention of a procedure or rule.

4 Basic concepts

4.1 General

This clause covers the concepts and vocabulary fundamental to the ISO/IEC/IEEE 15026 series.

4.2 Assurance

The ISO/IEC/IEEE 15026 series uses a specific definition for assurance as being grounds for justified confidence. Generally, stakeholders require grounds for justified confidence prior to depending on a system, especially a system involving complexity, novelty or technology with a history of problems (e.g. software). The greater the degree of dependence, the greater the demand for strong grounds for confidence. The appropriate valid arguments and evidence establish a rational basis for justified confidence in the relevant claims about the system's properties. These properties may include such aspects as future costs, behaviour and consequences. Throughout the life cycle, adequate grounds justify decisions related to ensuring the design and production of an adequate system and to be able to place reliance on that system.

Assurance is a term whose usage varies among the communities who use the term. However, all usage relates to placing limitations on or reducing uncertainty in such things as measurements, observations, estimations, predictions, information, inferences or effects of unknowns with the ultimate objective of either achieving or showing a claim, or both. Such a reduction in uncertainty can provide an improved basis for justified confidence. Even if the estimate of a property's value remains unchanged, the effort spent in reducing uncertainty about its value can often be cost-effective since the resulting reduced uncertainty improves the basis for decision-making.

Assurance justifies the confidence that:

- a) the system or software, as specified, meets real-world needs and expectations;
- b) the as-built and operated system meets the specifications; or
- c) both a) and b).

Specifications may be representations of either static or dynamic aspects of the system, or both. Specifications often include descriptions of capability, functionality, behaviour, structure, service and

responsibility including time-related and resource-related aspects as well as limitations on frequency or seriousness of deviations by the product and related uncertainties.

Specifications may be either prescriptions or constraints, or both (e.g. for and on product behaviours), and may as well include measures of merit and directions regarding trade-offs. Generally, specifications place some limitations on the environment and its conditions (e.g. temperature) and possibly the conditions of the product (e.g. age or amount of wear).

4.3 Stakeholders

Through their life cycle, systems and software have multiple stakeholders who have an interest in, affect, or are affected or impacted by, the system and the system life cycle processes. Stakeholders can benefit from, incur losses from, impose constraints on, have an interest, right, share, or claim in, or likely to be affected or impacted by, or otherwise have a “stake” in the system, and therefore are those that define the requirements for the system. Stakeholders can include the system’s owners and users, developers and operators who are required to identify requirements for the development and operation of the system, end users, persons or groups affected or impacted, organizations representing persons or groups affected or impacted, managers, acquirers, maintainers, regulators, taxpayers, certifying agencies, and markets.

A different but important kind of stakeholder is an attacker, who certainly imposes constraints or has interests involved with the system. This document includes the attacker as a stakeholder; however, some in the security community and elsewhere exclude attackers from their use of the term “stakeholders”.

Depending on conditions and consequences, the various stakeholders require grounds for justified confidence in properties of the system which affect or impact them.

Note There can be direct and indirect stakeholders. Indirect stakeholders are not directly affected or impacted by a system but are indirectly affected or impacted by it.

4.4 System and product

To be consistent with ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207, The ISO/IEC/IEEE 15026 series uses the term “system” throughout. Users of this document who are more familiar with using the term “product” should note that “system” includes products and services that are the results of processes as well as software and system or software elements or components. While primarily motivated by concern for systems produced (at least in part) by human-controlled or artificial processes, this document can be used in reducing uncertainty about a system’s dependence on natural phenomena as well.

4.5 Property

4.5.1 General

The ISO/IEC/IEEE 15026 series relates assurance to requirements of a property of a system or software product. Properties are entities that can be predicated of things or, in other words, attributed to them. Therefore, a property can be a condition, a characteristic, an attribute, a quality, a trait, a measurement or a consequence. A property can be invariant or dependent on time, situation or history. In the ISO/IEC/IEEE 15026 series, a property is expected to be relevant directly or indirectly to a system or systems and thus have related requirements.

Requirements can apply to values of properties, over time or at any specific point in time. Generally, the last is the most important in the ISO/IEC/IEEE 15026 series. As this knowledge involves predicting the future, it is often the most difficult and uncertain to attain; therefore, a system’s future behaviour and consequences (see 5.8) often become principal issues in its assurance.

Many of the properties with requirements are qualities of the system. Several standards and reports, including standards on Systems and software Quality Requirements and Evaluation (SQuaRE) developed by JTC 1/SC 7, ISO/IEC 2382 and ISO 9241, provide lists and definitions of qualities that can be the subject of assurance.

This use of the term “property” derives from, is consistent with and subsumes the broad use of the term “property” in ISO/IEC 25010.

Producers and other stakeholders can prioritize properties such as efficiency and reliability and perform trade-off studies between them and their related requirements. A number of techniques have been created for addressing these trades. The specifying of a top-level claim for a property is sometimes the result of analyses including trade-off studies.

4.5.2 Properties as behaviours

Often the property is specified as behaviour. During performed operations, behaviour-related properties can be formally specified as a combination of:

- restriction on allowed system states (sometimes called the “safety property”);
- system states that must be reached; required progress or accomplishment (“liveness property”);
- constraints on flows or interactions; requirements for separation constraint.

These kinds of properties can be stated as conditions or constraints that must be true of the system.¹⁾ In practice, these are non-trivial and modularized, involving time and starting state(s) as well as state transitions related to interaction with the system’s or software’s environment.

Many kinds of flows such as of gases, fluids, traffic or information are of possible interest as well as constraints on them such as non-interference and separations to be maintained. In addition, flow constraints are often convenient or necessary to specify aspects of information security including access control mechanisms and policies and restrictions on information overtly or covertly communicated.

4.6 Uncertainty and confidence

Uncertainty is used in the ISO/IEC/IEEE 15026 series as an inclusive term. It covers lack of certainty whether the uncertainty can be modelled probabilistically or not. Uncertainty can include vague notions modelled without the use of probability. Certain communities restrict the application of this term to predictions of future events, to physical measurements already made or to unknowns. While these limited usages can be convenient within those communities, users of the ISO/IEC/IEEE 15026 series span many communities.

The degree of confidence that can be or is justifiably engendered based on a specific assurance case can vary by individual or organization and the situation. The less uncertainty about an assurance case’s claims, the higher the degree of justified confidence. However, the conversion of an amount of uncertainty into a degree of justified confidence in suitability for certain applications is not straightforward or well understood. For this and other reasons, consequences are sometimes directly included within the assurance case. While this closes a logical gap, it does not remove the decision maker’s act of judgement regarding the merited degree of confidence.

4.7 Conditions and initiating events

The assurance case covers all the conditions that can have a significant negative effect on the conclusion and uncertainty of the top-level claim. The potentially relevant universe of conditions and events can be hard to initially identify and ascertaining which ones have a significant effect can be difficult without at least initially including them in the assurance case.

Historically, the one condition that has received the most attention is system failure. A substantial volume of checklists, practice and literature exists concerning system failure (e.g. IEC 61508-7). While much of this work has been done in the communities addressing safety, security or human error, system failure can result in less achievement of a positive property or consequence as well as negative properties or losses.

The dangerousness of system behaviours can differ by the conditions of its environment. These behaviours and conditions often are combined during analysis to establish whether adverse consequences can result or

1) If specified formally, this can allow static analysis of conformity of designs and code, potentially adding creditable assurance evidence.

not. The actual conditions of its environment is known or is not known within the system depending on its sensors or inputs and their processing.

The designers of the system are or are not cognizant of all the initiating events for a condition within the environment; however, dangerous conditions can be dealt with even though not all of their initiating events are recognized or recognizable.

4.8 Consequences

Outside the system, much of the reasoning is based on conditions that can lead to adverse consequences and their initiating events or preconditions. Inside the system reasoning is based on conditions that can lead to dangerous system behaviours and the initiating events or preconditions for these conditions.

In practice, claims can extend beyond the boundaries of the system or its behaviours. In particular, claims can place limitations on consequences of a system's behaviour or system-related events, activities, or conditions, or all – especially on the values of consequences. A consequence is desirable or undesirable from a stakeholder's perspective, viewpoint or interests. A consequence can occur anywhere in the system's life cycle.

In complex socio-technical systems, explanations of mishaps or claim violations cannot be limited to "component" failures. Adverse consequences can result from normal behaviour variability and unintended or unanticipated interactions. Regardless of how they arise, dangerous conditions and adverse consequences are subjects for mitigation.

Attackers can possess capabilities, resources, motivations and intentions that enable them to initiate and carry malicious efforts to violate a claim. Violators use their capabilities to take advantage of either system-provided or environment-provided opportunities, or both, called vulnerabilities, i.e. weaknesses that can be exploited or triggered by a threat source²⁾.

A sometimes-misunderstood point is that maliciousness and subversion are concerns even when no security-related system property is involved. Malicious developers can have an effect on successful achievement of almost any property.

Several standards or reports mention consequences associated with systems within a specific domain. Examples include the ISO 14620 series and ISO 19706. Risk management standards also address consequences, for example, ISO/IEC/IEEE 16085 and ISO 31000.

5 Using multiple parts of the ISO/IEC/IEEE 15026 series

5.1 General

The ISO/IEC/IEEE 15026 series or its parts can be used alone or with other standards or guidance. The parts of the ISO/IEC/IEEE 15026 series can be mapped to most life cycle standards and can use any set of well-defined qualities or properties.

5.2 Initial usage guidance

The properties or claims, or both, covered when using the ISO/IEC/IEEE 15026 series are entirely up to the users of the standard who are responding to the system's stakeholders' needs and requirements. Any property or claim can be selected for an assurance case, regardless of its importance or related risk; however, the parts of the ISO/IEC/IEEE 15026 series are designed primarily for those properties that one or more primary stakeholders deem critical. ISO/IEC/IEEE 15026-4 uses the term "critical properties" for these stakeholder priorities and requirements.

ISO/IEC/IEEE 15026-3 opens up new engineering and decision options, because it takes not only a standalone perspective but also one that includes relating integrity levels to an assurance case. ISO/IEC/IEEE 15026-3

2) For many purposes, the meaningfulness and desire to separate vulnerabilities from other weaknesses can be low or non-existent. In addition, a question always exists about the current and future contexts that are relevant for "can be exploited or triggered".

concentrates more on the system itself and its integrity levels rather than on external risk analysis and also includes the creation of integrity levels. [Clause 6](#) discusses integrity levels.

5.3 Relationships among parts of the ISO/IEC/IEEE 15026 series

If the risks or the risk treatment are not well understood or if the dependency structure of the whole system or the choice of suitable claims is unclear, using an assurance case is a better choice than using integrity levels. This particularly is the case when facing new kinds of risks or using a new kind of risk treatment. In these situations, justifying the choice of the top-level claim for the assurance case is important.

When the risks and their treatment are well understood, however, developers can select the proper claims for their context from a known set without justifying the choice of top-level claim — an integrity level from a set of integrity levels. In these situations, the generic arguments created by the definers of the integrity level provide the justification that meeting the integrity level requirements can adequately show the meeting of the integrity level. Such a justification (e.g. a generalized assurance case) is usually created one time by a separate organization and used by multiple projects.

5.4 Authorities

Parts of the ISO/IEC/IEEE 15026 series involve authorities. For example, ISO/IEC/IEEE 15026-3 includes obtaining agreements among the integrity level definition authority, the design authority and the integrity level assurance authority. Additionally, a new system typically has approval authorities of acquirers that take charge of analysing the process of creating assurance cases with the design authority and the integrity level assurance authority of the suppliers.

However, the “approval authority” for the assurance case is not necessarily the judge of conformity to ISO/IEC/IEEE 15026. To the extent possible, claims of conformity to ISO/IEC/IEEE 15026 are judged on aspects that are more straightforward and more difficult to dispute than the quality of artefacts and decisions judged in the context of the system or project. In practice, contracts can explicitly call for the acquirer to be the approval authority or the approver of conformity to ISO/IEC/IEEE 15026.

6 The ISO/IEC/IEEE 15026 series and the assurance case

6.1 General

ISO/IEC/IEEE 15026-2 covers the structure and content of an assurance case. It describes the five principal components of an assurance case: claims, arguments, evidence, justifications and assumptions. The purpose of an assurance case is to improve assurance communications by informing stakeholders’ decision-making and supplying grounds for stakeholder confidence. The most common use of an assurance case is to provide assurance about system properties to parties not closely involved in the system’s technical development processes. Such parties can be involved in the system’s certification or regulation, acquisition or audit. Usually, an assurance case addresses the reasons to expect and confirm successful production of the system, including the possibilities and risks identified as difficulties or obstacles to developing and sustaining that system.

Unlike logical proofs of the deduction of the claims from the evidence, which covers the absolute truth or Platonic truth aspects, assurance cases deal with the dialectic aspects of the system where the truth is always relative or even subjective. In other words, logical proofs are described under a fixed logical theory, but assurance cases can be rebutted on the basis that the underlying logical theory is inappropriate. Assurance cases arise when one realizes the properties of the systems in the real world can never be completely formalized in a logical theory, but there is always something which is not covered by any logical formalization.

NOTE 1 When the top-level claim is about safety, security, dependability or RAM (reliability, availability and maintainability), assurance cases associated with these claims are called safety cases, security cases, dependability cases or RAM cases, respectively.

Considered as an artefact, an assurance case has quality-related aspects such as: the nature of content, its form or structure (e.g. method of argumentation or modularity); semantic issues such as completeness; creation and maintenance (including tool support, usability and presentation, integrity, validity,

understandability); and the feature of clearly stated conclusions with explicit degrees of uncertainty. The quality-related aspects of an assurance case are not covered in the ISO/IEC/IEEE 15026 series.

Any substantive modifications in the system, changes in the environment or changes in the assurance case's top-level claims necessitate recorded changes to the assurance case. Thus, an assurance case usually contains a progressively expanding body of evidence built up during development and later life cycle activities that responds as required to all relevant changes.

NOTE 2 An assurance case's claim(s) on the values of properties can include the system's entire set of requirements for a property of interest. One example can have a top-level claim composed of:

- a) required limitations on consequences;
- b) functionality and properties of the system itself (e.g. that this functionality cannot be bypassed).

The qualities defined in the standards on Systems and software Quality Requirements and Evaluation (SQuaRE) developed by JTC 1/SC 7 include qualities related to functionality and constraints.

6.2 Justification of method of reasoning

An argument has an associated justification for the validity or merit of its method of reasoning. The method of argument can be an additional source of uncertainty.

A variety of bases for argumentation and analysis in the assurance case can be used, and these vary in their applicability, power, resulting accuracy and uncertainty and ease of use. Subjects of and approaches to reasoning differ among communities having differing motivations, mind-sets and often multiple methods of reasoning.

Examples of methods of reasoning include:

- quantitative:
 - deterministic (e.g. formal proofs);
 - non-deterministic formal systems for reasoning:
 - probabilistic;
 - game theoretic (e.g. minimax); or
 - other uncertainty-based formal systems of reasoning (e.g. fuzzy sets);
- qualitative (e.g. staff performance evaluations, court judgements and qualitative statements of event causality).

Complex products and situations — and any involving humans — are beyond the current state of the art to “quantitatively” create precise and accurate predictions. Subjective judgements are used in the absence of affordable, suitable and more objective methods and techniques or to supplement or evaluate the results of such techniques. Supplementing quantitative techniques with expert review and judgement is widely used and generally accepted. As with other forms of argumentation, subjective judgements take the form of a claim and its support. While sometimes necessary or advantageous, use of subjective judgement within the assurance case can lead to additional uncertainties, so, generally (just as with assumptions), the less critical the judgement is, the better.

The patterns of occurrences of “natural” events and common, non-malicious human behaviours are usually described probabilistically. However, possibilities for intelligent, malicious actions whose probability is not determinable or not knowable is particularly a concern if the intelligent, malicious adversary deliberately violates any probability estimates one can make regarding their behaviour, e.g. to achieve surprise. This distinction is central to the difference in reasoning between safety and security.

6.3 Means of obtaining and managing evidence

For any property, many means of obtaining evidence exist. Among these are human experience, history, observations, measurements, tests, evaluation and compliance results, analyses, defects and inferences. The evidence should achieve the objectives claimed in the assurance argument.

The body of evidence can become quite large and can be organized and managed by some framework providing permanence and traceability of the evidence in order to provide users confidence in its source, contents and validity. One guidebook indicates the following.

- Evidence should be uniquely identified so that arguments can uniquely reference the evidence.
- Evidence should be verifiable and auditable.
- Evidence should be protected and controlled by configuration management.
- Evidence should be accompanied by the metadata to properly use it within the assurance case.

Evidence here includes artefacts and reports concerning assurance, so they should be protected and controlled by configuration management. The last bullet is simply a restatement of what testing is supposed to achieve related to the assurance case.

6.4 Certifications and accreditations

Every aspect having potential significant consequences for meeting the top-level claim or for the confidence of key stakeholders has a potential place in the full assurance case evidence. Assurance case evidence should not only give coherent confidence to stakeholders, but also contain enough information to be used by certifiers and accreditors.

NOTE 1 The aviation and nuclear power industries have long histories of standards relating to certifications, and the security community in ISO/IEC JTC 1/SC 27 has been working on the topic of assurance for many years. Security examples include the Common Criteria, FIPS 140 for cryptology, and ISO/IEC 27002, combined with ISO/IEC 27001 (formerly with UK standard BS 7799-2:2002) form a basis for an Information Security Management System (ISMS) certification of an operational system. The UK Ministry of Defence and Civil Aviation Authority have also produced standards of interest including assurance-case-based standards for reliability, maintainability and safety.

NOTE 2 The safety community (e.g. commercial aviation) has used certification (designated agent or licensure) of key personnel as part of its approaches. A number of safety and computer security certifications exist from management-oriented ones to technical ones about specific products.

7 The ISO/IEC/IEEE 15026 series and integrity levels

7.1 General

Integrity levels are suitable for use for certain levels of risk, can support an assurance case, and impose criteria especially on the project, evidence collected and system. An integrity level can be viewed as a representation of the degree of confidence that is used to reach agreement among stakeholders of a system about risks related to that system.

ISO/IEC/IEEE 15026-3 first establishes an integrity level framework. The remainder of the standard covers defining integrity levels, using integrity levels, determining system or product integrity levels using risk analyses, assigning system element integrity levels, meeting integrity level requirements using evidence, and agreements and approvals involving authorities (see [5.4](#)).

Integrity level requirements reflect what is required to achieve and show that the system or system element has (or had or will have) the properties claimed by its integrity level. A system's integrity level states what is adequate in terms of properties of the entire system. Thus, showing the properties has a basic role in showing the meeting of larger claims involving the system and its environment including desirable or undesirable consequences. If such larger claims are not made, achieving and showing system element integrity levels supplies a basic part of showing the top-level claim regarding the system itself.

In practice, integrity levels are often discussed in terms emphasizing the evidence to meet the integrity level requirements and thereby provide evidence for the arguments supporting claims regarding properties of the system itself. However, the quality of the arguments justifying meeting integrity level requirements as showing the achievement of its related integrity level is also important because of the effect of that quality on uncertainties. Argument-, evidence- and assumption-related uncertainties are a part of establishing integrity level requirements.

NOTE Integrity levels and standards utilizing them have a significant history especially in safety. Integrity levels in safety-related standards are defined in multi-level sets addressing varying degrees of either stringency or uncertainty, or both, of their achievement with higher levels providing higher stringency and lower uncertainty. One example safety standard is the IEC 61508 series. Elsewhere, similar schemes are used with different labels, e.g. “conformance classes”.

7.2 Risk analysis

Risk analysis establishes the required integrity level for the entire system. Risk analysis is an ongoing and iterative process that should balance what is not yet knowable with what is required to be known. The integrity levels resulting from risk analysis are a translation of the values of consequences into the occurrences and timings of conditions or behaviours of the system. This translation is propagated to the integrity levels internal to the system and of its dependences as they are also in terms of occurrences and timings. Thus, integrity levels are a codification of what should be done and shown for various ranges and severities of limitations on property values and their associated uncertainties.

ISO/IEC/IEEE 15026 does not cover risk analysis in detail. Many standards and guidance documents exist that offer guidelines for risk analysis and can aid in the identification of potential adverse consequences. The IEC 61508 series and IEC 31010 provide approaches for risk analysis. As safety-specific terminology is used in IEC 31010, the terms “hazard” and “harm” should be interpreted as “dangerous condition” and “adverse consequence” respectively. The IEC 60300 series also provides guidance.

Other specialized standards include the ISO 13849 series on machinery, the ISO 14620 series on space systems, ISO 19706 on fire, ISO/TS 25238 on health informatics, ISO/IEC 27005 on information security and UK CAP 760 on air traffic and airports. Also of possible interest are the more general risk management standards ISO/IEC/IEEE 16085 and ISO 31000.

8 The ISO/IEC/IEEE 15026 series and the life cycle

8.1 General

ISO/IEC/IEEE 15026-4 provides a process view for systems and software assurance by providing a statement of purpose and a set of outcomes suitable for systems and software assurance. The concept of a process view is formulated and described in ISO/IEC/IEEE 24774. Like a process, the description of a process view includes a statement of purpose and outcomes. Unlike a process, the description of a process view does not include activities and tasks. Instead, the description includes guidance and recommendations explaining how the outcomes can be achieved by employing the activities and tasks of the various processes in ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207.

The system life cycle processes are provided in ISO/IEC/IEEE 15288 and software life cycle processes in ISO/IEC/IEEE 12207.

In the context of a life cycle model, the processes, activities and tasks shall be performed; guidance and recommendations of process views shall be followed. The ISO/IEC/IEEE 24748 series is intended to facilitate the joint usage of the process content of the two life cycle process standards. The ISO/IEC/IEEE 24748 series provides unified and consolidated guidance on the life cycle management of systems and software. Its purpose is to help ensure consistency in system concepts and life cycle concepts, models, stages, processes, process application, iteration and recursion of processes during the life cycle, key points of view, adaptation and use in various domains. ISO/IEC/IEEE 24748-1 illustrates the use of a life cycle model for systems in the context of ISO/IEC/IEEE 15288 and provides a corresponding illustration of the use of a life cycle model for software in the context of ISO/IEC/IEEE 12207.

ISO/IEC/IEEE 15026-4 gives the user the freedom to choose whether they use a specific artefact called an “assurance case” or document the assurance-related information in other documents. The point is to achieve the top-level claim and then to show the achievement of the claim for the value of a critical property for a relevant stakeholder. Life cycle processes, activities and tasks should reflect both realizing an adequate system and being sure that the system is adequate by showing that achievement to the required confidence of the stakeholders.

Users of ISO/IEC/IEEE 15026-4 can require risk assessment and risk management, measurement and requirements processes that are more fully elaborated than the treatments provided in ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207. ISO/IEC/IEEE 16085, ISO/IEC/IEEE 15939 and ISO/IEC/IEEE 29148 are designed to be used with ISO/IEC/IEEE 15288 and ISO/IEC/IEEE 12207 to provide more detail for these three processes. Other standards that provide useful requirements and guidance for selected processes are ISO/IEC/IEEE 15289 for documentation resulting from the execution of life cycle processes and ISO/IEC/IEEE 16326 for the project management process.

The ISO/IEC/IEEE 15026 series is intended to be compatible with these life cycle process standards. The goals of assurance, the selection of claims to be assured, assurance-related planning and the construction and maintenance of the assurance case have influences within all life cycle processes.

8.2 Assurance activities in the life cycle

The execution of a planned and systematic set of assurance activities should provide grounds for confidence in system properties. These activities are designed to ensure that both processes and systems conform to their requirements, standards and guidance and defined procedures. “Processes” in this context include all of the activities involved in the design, development and sustainment of the system. For software, “software products” include the software itself, the data associated with it, its documentation and supporting and reporting paperwork produced as part of the software process (e.g. test results and assurance arguments) to complete the assurance case. The “requirements” include requirements for the properties that should be exhibited, ultimately based on requirements to limit, reduce or manage property-related costs and losses. The “standards and guidance” can be technical, defining the technologies that can be used in the system or software, or they can be non-technical, defining aspects of the process that are further delineated by the “procedures” that make satisfaction of the system’s requirements possible.

Management of life cycle activities includes handling both the activities directly involving the assurance-related information and the effect that the assurance-related information has on other activities. This management is best performed when the top-level claims are considered from the beginning of concept development, used to influence all activities and systems, and became an integral part of the overall engineering process. These activities can all be done only if the system and the body of information showing achievement of those claims were being developed concurrently.

This parallel nature of development rationale and argument is but one of the advantages of concurrent development of the system and its assurance case. The development process and the system can be aimed not only at achieving the claim but doing so in a way that can be shown to be adequate by the assurance case. The assurance case influences the system by causing it to be developed in such a way that an argument is more practical to construct. This often results in a simpler system (at least internally), a system whose system elements can be used in isolation to show certain sub-claims, and an arrangement of system elements such that reasoning about the composition is both within the state of the art and practical. Concurrent processes can include requirements covering more conditions and events as well as adequate resilience, methods being used that produce few faults and validation or verification being targeted to what should be shown and showing that adequately^[110].

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